

Localization and Object Detection

From one label to a variable set of boxes

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

What task are we solving?

Where we are

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Classification asks **what is in the image?** — a single label. But most real images contain **several** things, each **somewhere**. This lecture is about **what** and **where**, for a variable number of objects.

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Detection predicts a **variable set of objects**,
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the output is no longer a fixed-size vector — it is a **set** whose size depends on the image.

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- the whole image is assumed to be **about one thing**.

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classification + a box regression head

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- this variable N is what makes detection **fundamentally harder**.

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Segmentation pushes further: a class label for **every pixel**, not just a box. That is **Lecture 10**. Today we stop at boxes.

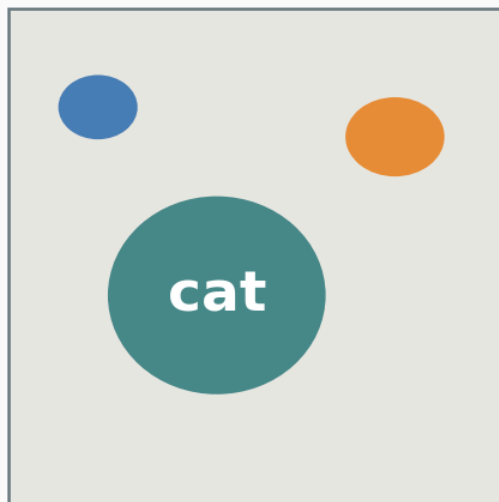
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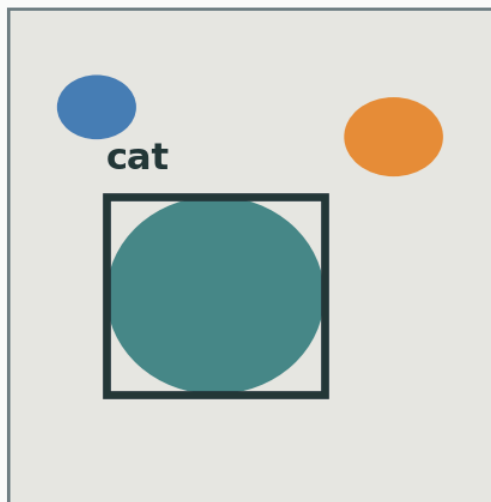
Task	Output	Granularity
classification	$\hat{y}$	whole image
localization	$(\hat{y}, \hat{b})$	one object
detection	$\{(\hat{y}_i, \hat{b}_i, s_i)\}$	many objects
segmentation	pixel labels	every pixel (L10)

The same image, three tasks

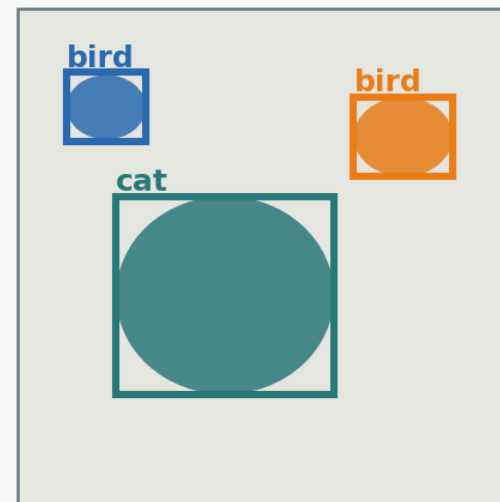
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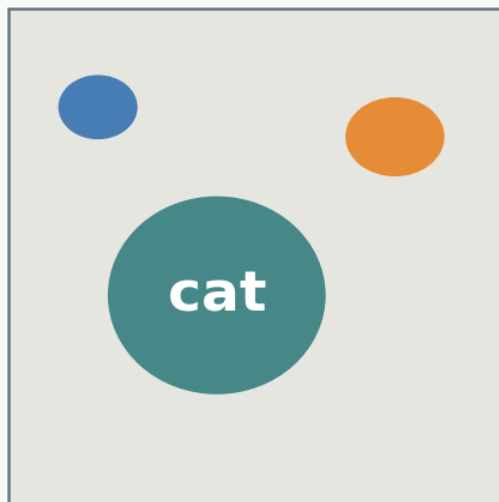


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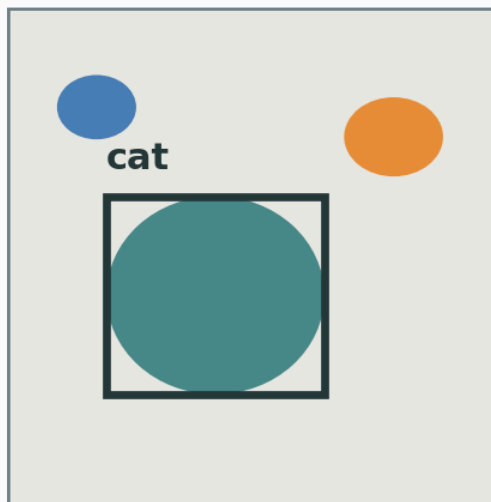


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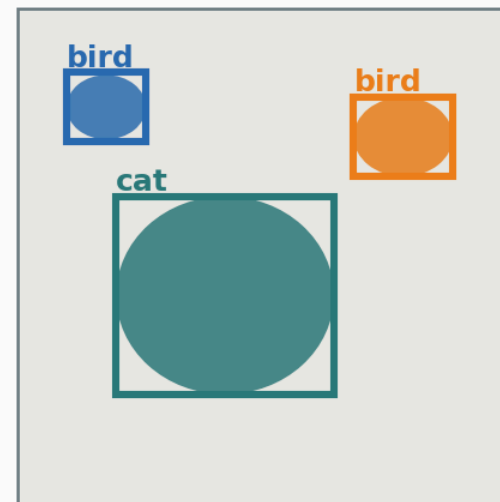
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one label · one box · a **set** of boxes — the output grows richer left to right.

Bounding boxes and localization

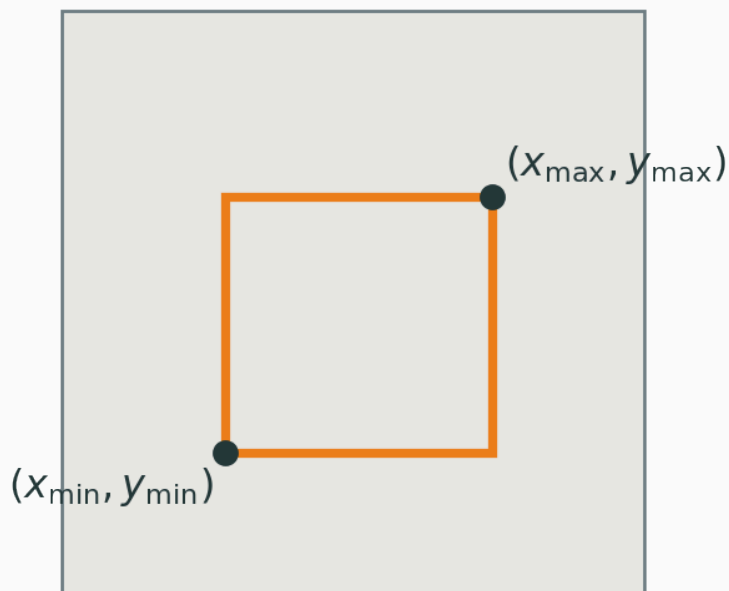
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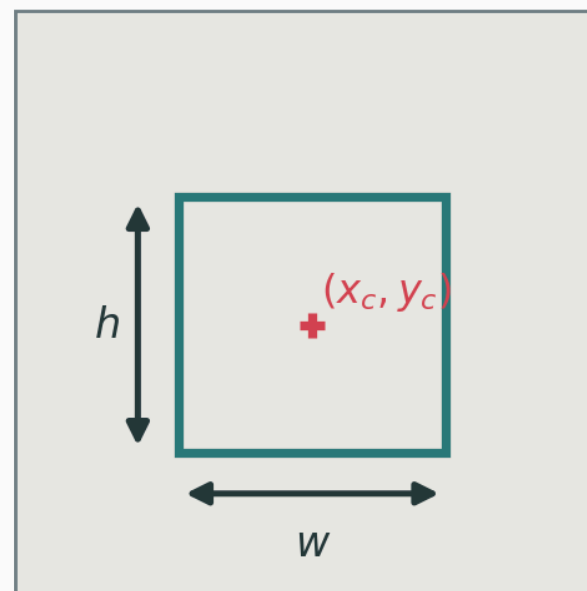
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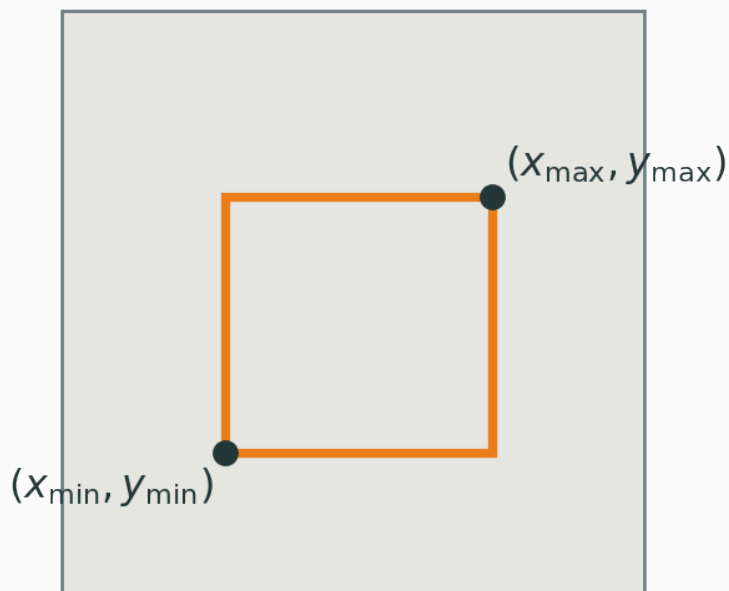
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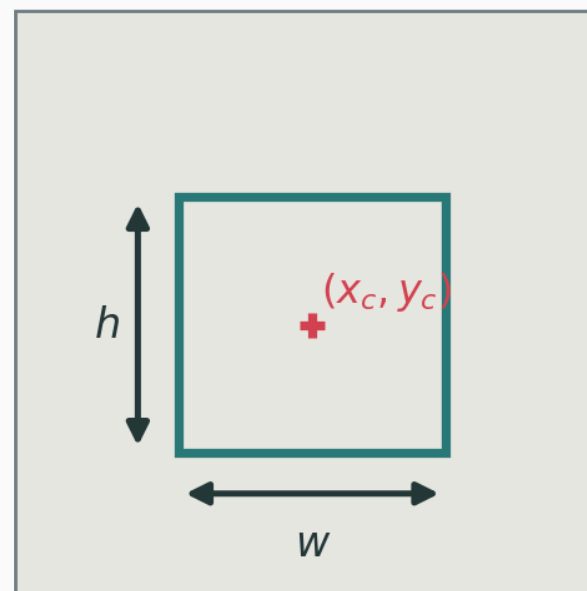
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$$b = (x_{\min}, y_{\min}, x_{\max}, y_{\max}) \quad \text{or} \quad b = (x_c, y_c, w, h)$$

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Normalized coordinates live in $[0, 1]$ regardless of resolution — so the same network handles images of **any** size, and the regression targets are **well-scaled**.

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one backbone, two heads — a class distribution and four box numbers

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This **multi-task loss** — a shared trunk with several weighted heads — recurs throughout detection.

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L_2 punishes large errors hard (sensitive to outliers); L_1 is steadier but kinks at 0.

Smooth L_1 : the best of both

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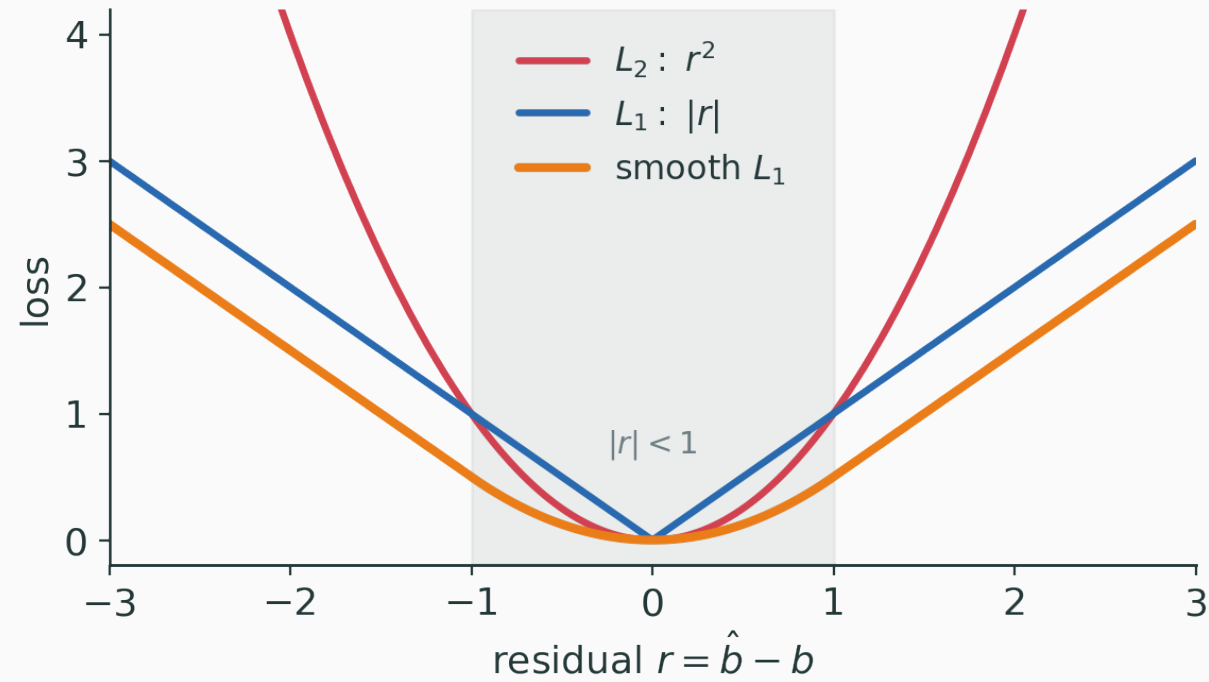
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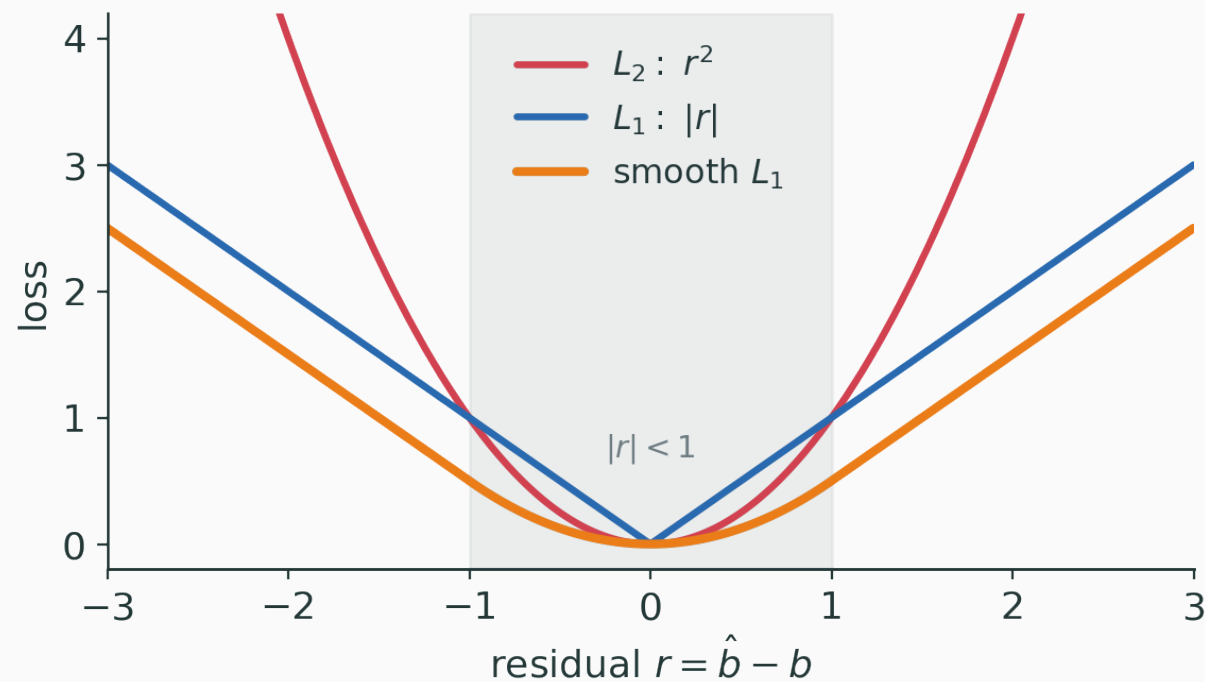
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$r = \hat{b}_j - b_j$ is the per-coordinate residual — the pieces meet smoothly at $|r| = 1$.





near 0 it curves like L_2 (no kink) · far out it grows like L_1 (no outlier blow-up).

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$$L_1 = 0.35$$

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/bbox-loss

drag the predicted box around a fixed ground-truth box and watch L_1 , L_2 , and smooth- L_1 update coordinate-by-coordinate, and see where each loss's gradient is large or flat.

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Q. Coordinate losses treat all four numbers independently. What about the box do they **not** directly measure?

IoU and detection metrics

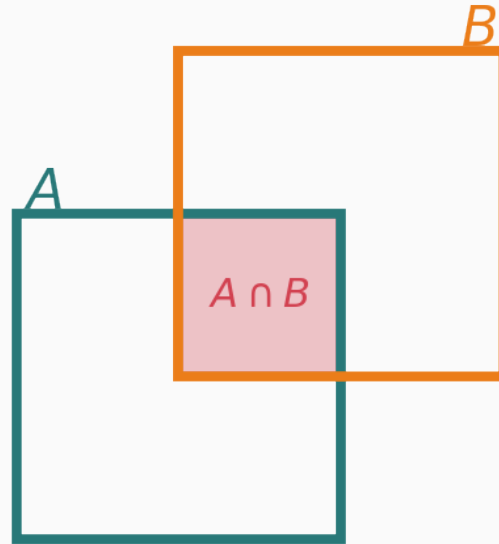
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$$|A| = 100, \quad |B| = 100, \quad |A \cap B| = 25$$
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an overlap of 25 on boxes of area 100 still scores only 0.14 – IoU is a **strict** measure.

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so: optimize an IoU-based loss directly

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If the boxes **do not overlap**, $\text{IoU} = 0$ for **all** non-overlapping configurations. The loss is flat — **zero gradient** — so it cannot pull a far-off box toward the target.

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later variants (DIoU, CIoU) refine this further — all share the same fix: keep a signal when IoU is 0.

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Precision and recall

To **score a detector**, first count matches (a prediction is a **hit** if IoU with a true box $> \tau$):

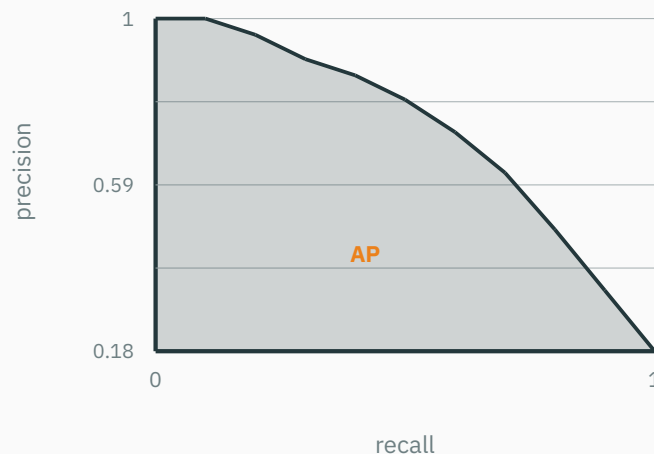
$$\text{precision} = \frac{TP}{TP + FP} \quad \text{recall} = \frac{TP}{TP + FN}$$

- **precision** — of the boxes I predicted, how many were **right**?
- **recall** — of the objects that exist, how many did I **find**?

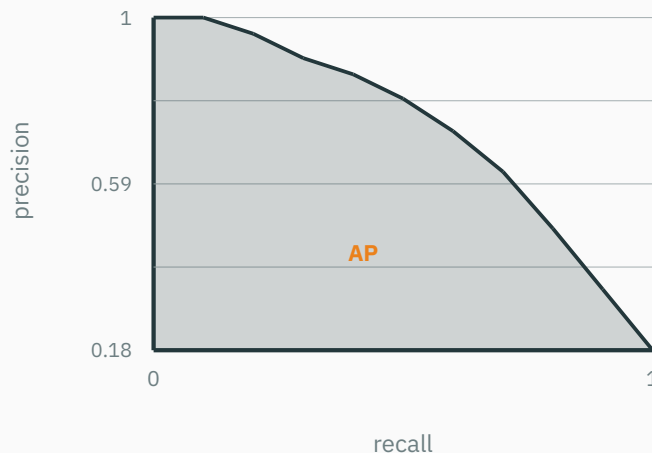
Lowering the score threshold finds more objects ($\uparrow$ recall) but admits more false alarms ($\downarrow$ precision). There is a **tradeoff curve**.

Sweep the score threshold; plot precision vs recall; AP is the **area under it**:

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$$\text{AP} = \int_0^1 \text{precision}(r) dr \in [0, 1]$$

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mAP — one number summarizing detection quality

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/object-detection

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Q. A detector reports every object with **very low** confidence. What happens to its precision, and to its recall?

Building a detector

Why detection is hard

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Every design choice below is a response to one of these four problems.

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turn a variable-size set problem into a **fixed, dense prediction problem**

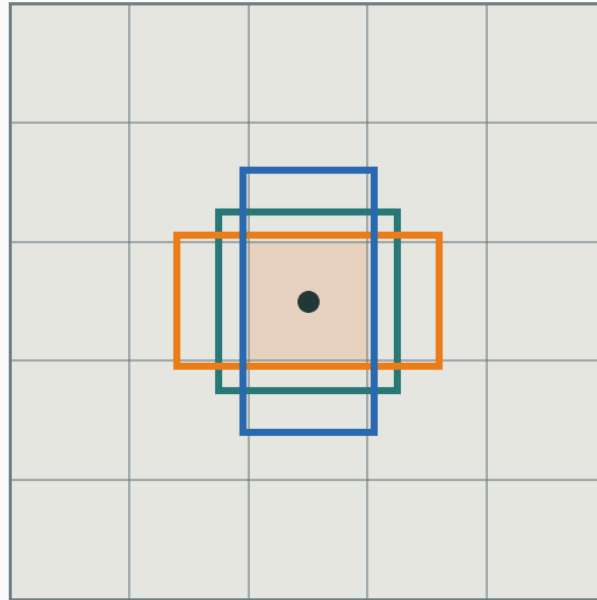
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the network **adjusts** an anchor rather than inventing a box from scratch.

The network outputs **corrections** to each anchor — easier to learn and well-scaled:

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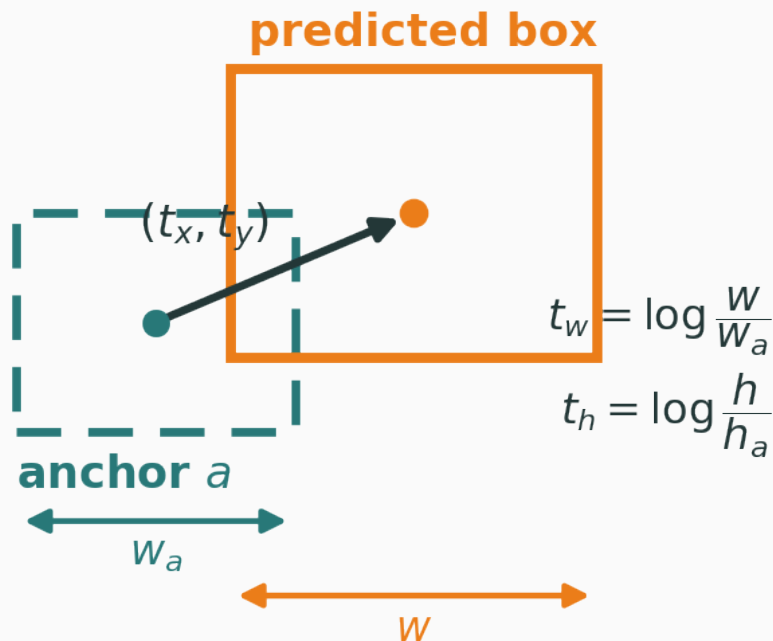
$$t_x = \frac{x - x_a}{w_a}$$

$$t_y = \frac{y - y_a}{h_a}$$

$$t_w = \log\left(\frac{w}{w_a}\right)$$

$$t_h = \log\left(\frac{h}{h_a}\right)$$

centers in anchor-widths, sizes in log-scale; **all-zeros** reproduces the anchor exactly.



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all-zero offsets would reproduce the anchor exactly; here the box shifts right, up, and doubles in width.

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for A anchors and K classes: $A \times (1 + K + 4)$ numbers per location.

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- a positive anchor's box/class targets come from the matched b_{gt} ;
- for a simple RPN-style rule, common teaching values are $\tau_+ \approx 0.7$, $\tau_- \approx 0.3$; modern detectors may use task-specific or learned assignment instead.

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foreground : background can be **1 : 1000s** — the loss needs to **ignore the easy majority**.

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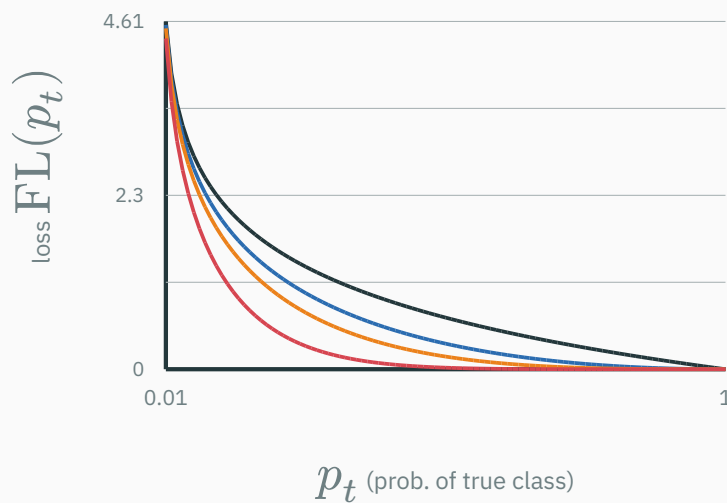
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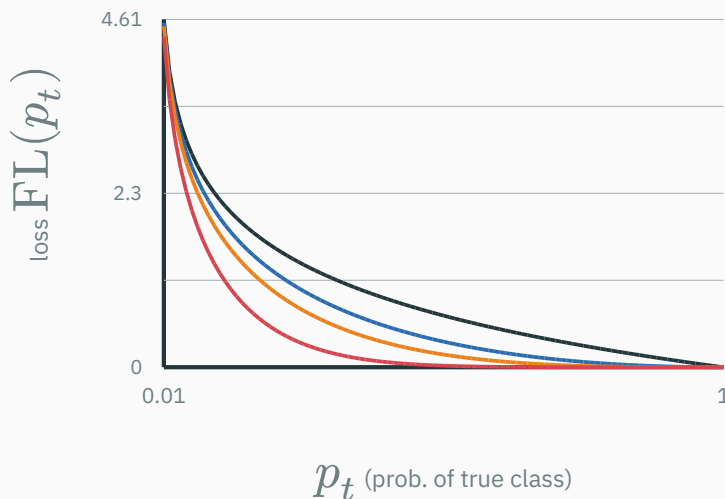
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numeric ($\gamma = 2$): an easy $p_t = 0.9$ scales its loss by $(1 - 0.9)^2 = 0.01$ — just **1%** of plain CE.

Focal loss down-weights easy examples



$\gamma = 0$ (CE) $\gamma = 1$ $\gamma = 2$ $\gamma = 5$



$$\gamma = 0 \text{ (CE)} \quad \gamma = 1 \quad \gamma = 2 \quad \gamma = 5$$

higher $\gamma \Rightarrow$ easy examples ($p_t \rightarrow 1$) contribute almost nothing — built for **dense one-stage** detectors.

INTERACTIVE

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Q. If you set τ_+ very high (say 0.9), what happens to the number of positive anchors — and to training?

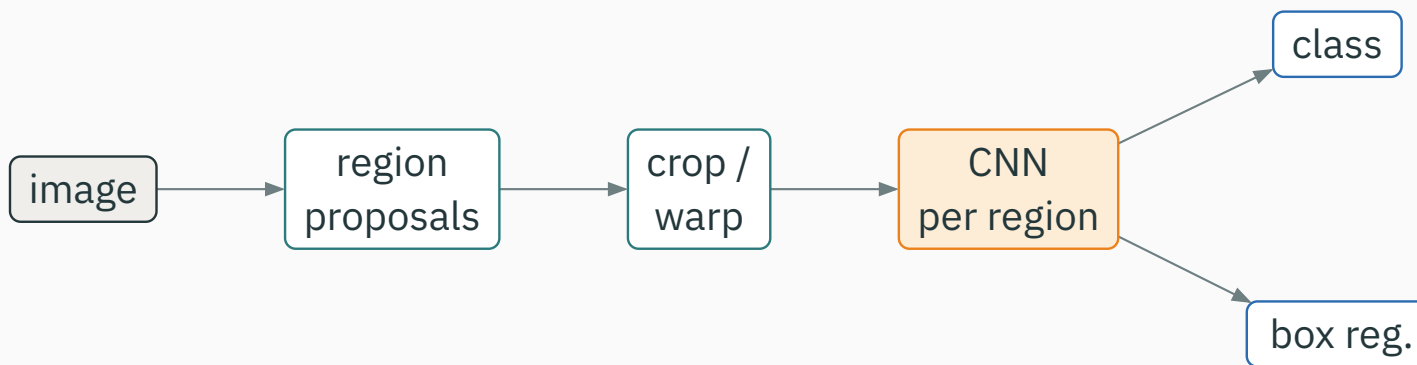
Two-stage detectors: the R-CNN family

R-CNN: propose, then classify

Stage 1 proposes regions (classic **selective search**); stage 2 runs a CNN on **each**.

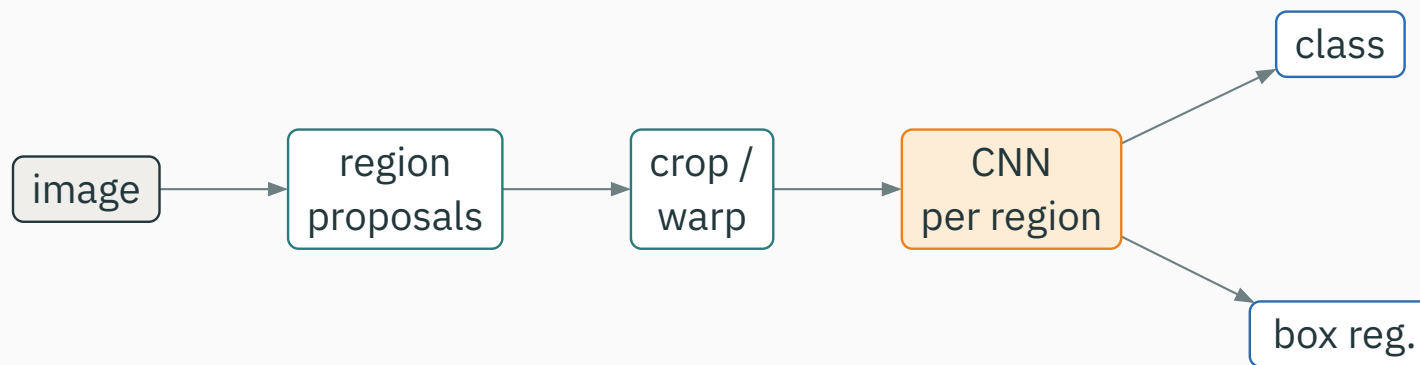
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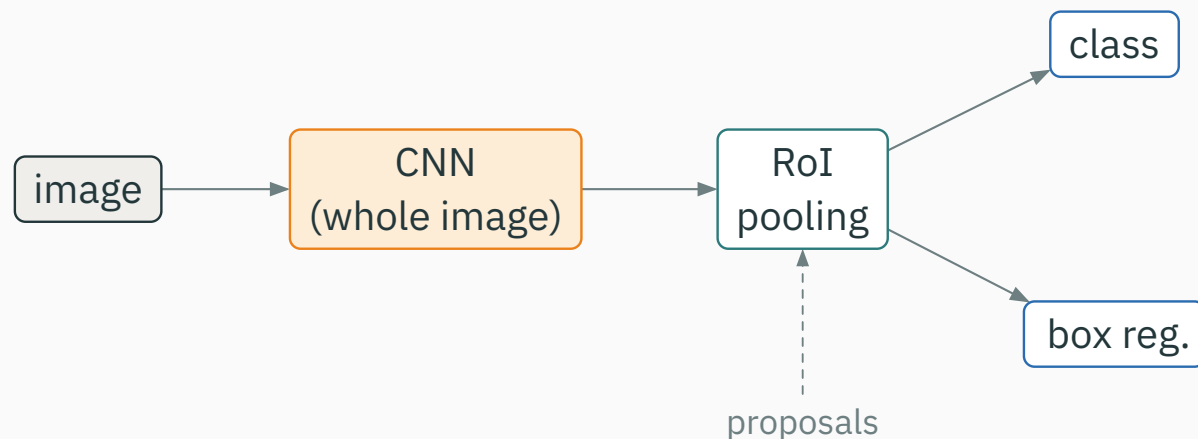
A CNN forward pass **per region** (thousands of crops) — **slow**, and the multi-stage training (proposals, CNN, SVMs, regressors) is cumbersome.

Fast R-CNN: share the convolution

Run the CNN **once** over the whole image; pool features **per proposal** (**RoI pooling**).

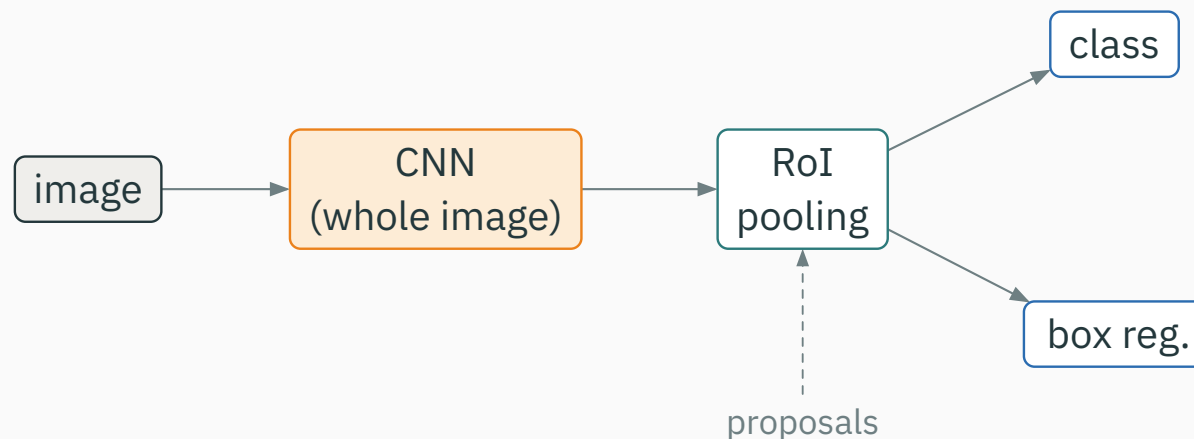
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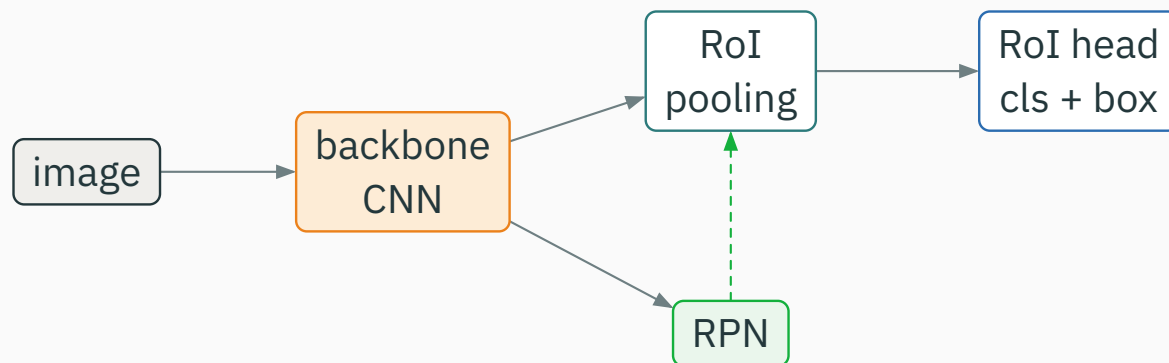
The expensive convolutions are computed **once** and **shared** across all proposals — a large speedup, and the heads train **end-to-end**.

Faster R-CNN: learn the proposals too

Replace hand-crafted proposals with a **Region Proposal Network** (RPN) on the same features.

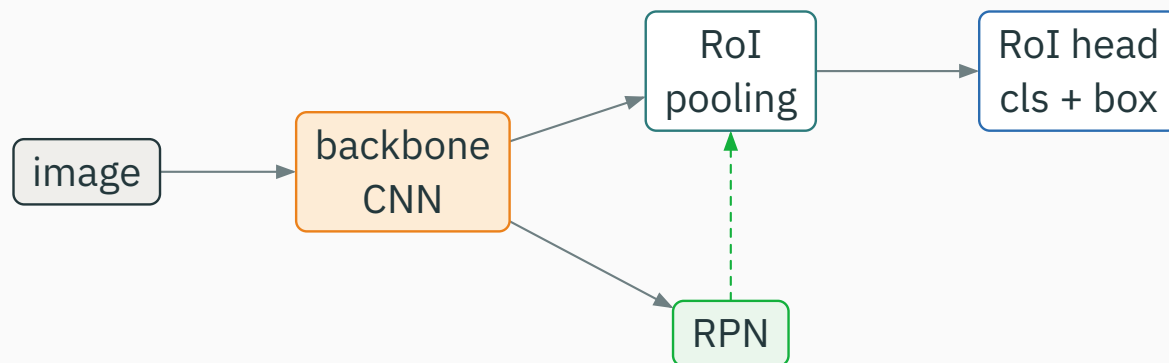
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the RPN and the detection head **share the backbone** — proposals become **neural**, fast, and trainable.

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trained jointly – one network, two stages, four loss terms

The two-stage intuition

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Stage 1 casts a wide net (high recall, coarse); stage 2 is precise (class + tight box). **Historically** this two-stage split gave strong localization.

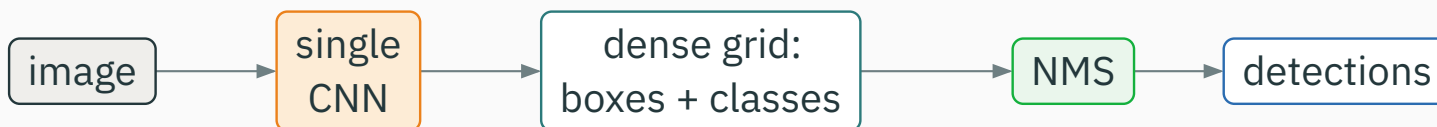
One-stage detectors and NMS

YOLO: detection in a single pass

You Only Look Once — one network maps the image **directly** to boxes + classes.

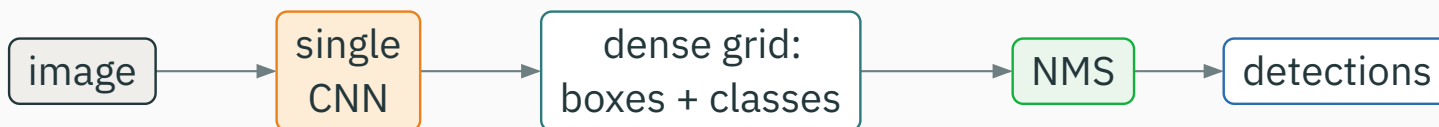
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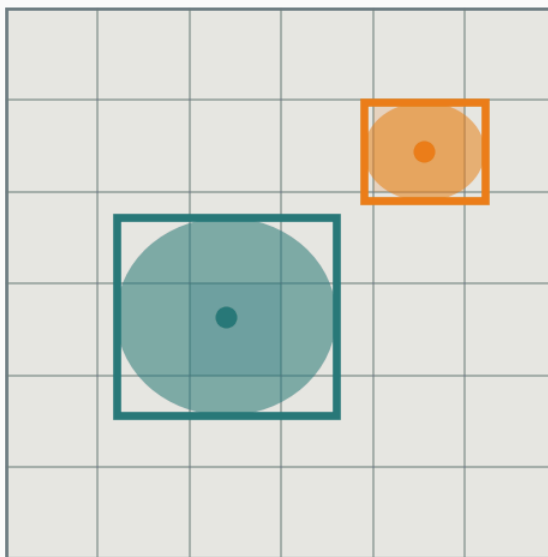


No separate proposal stage: a **single** forward pass regresses all boxes and classes at once, then NMS removes duplicates. Simple and **fast**.

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the cell containing an object's **center** is responsible for predicting it.

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same ingredients as Faster R-CNN — just predicted **all at once**, no region-proposal stage.

Speed vs accuracy — carefully

	one-stage (YOLO)	two-stage (Faster R-CNN)
passes	single	propose, then refine
speed	faster, simpler	heavier
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These are **historical** tendencies. Modern one-stage detectors (with focal loss, better assignment, stronger backbones) **close much of the gap** — the real ranking depends on implementation, data, and scale.

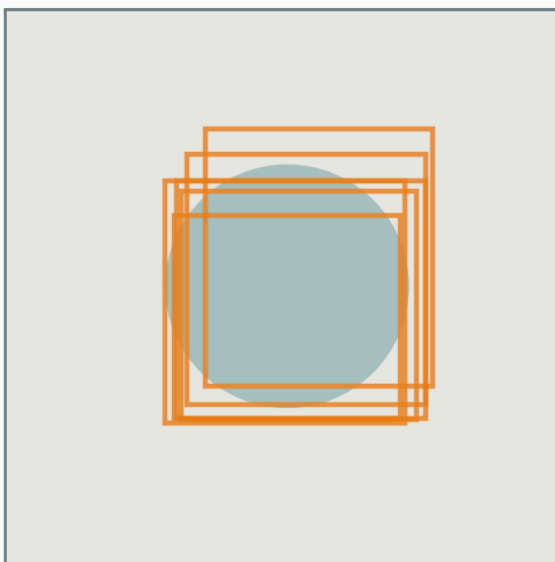
The duplicate problem

The dense grid fires **many** overlapping boxes for one object:

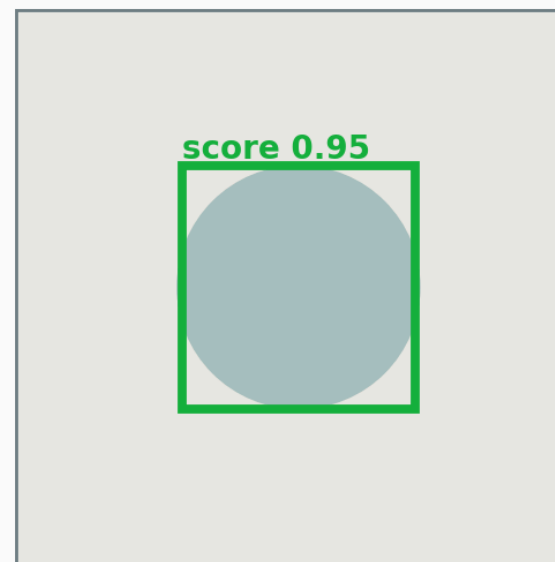
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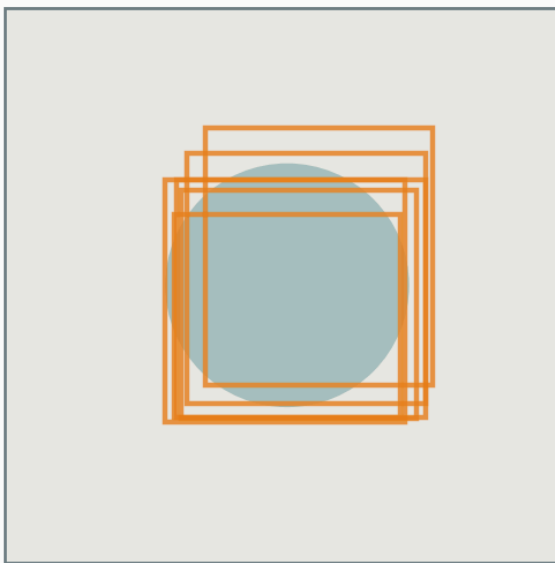
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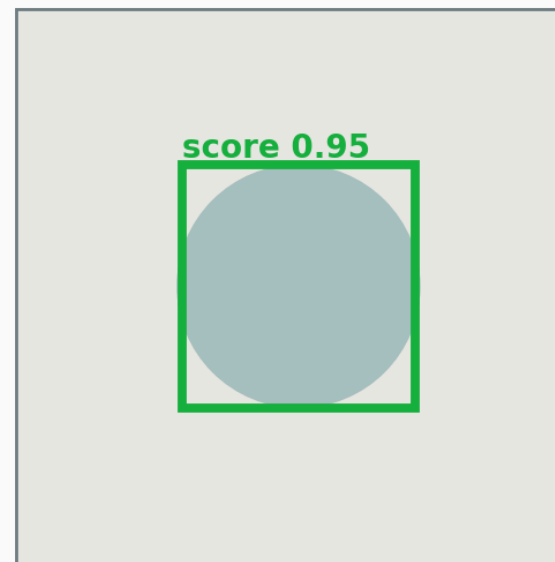
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before NMS: many boxes / object



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we want **one** box per object — a post-processing step must **suppress the duplicates**.

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Each kept box “claims” its neighbourhood; overlapping lower-score boxes are discarded. Typical $\tau \approx 0.5$. (Soft-NMS **decays** scores instead of hard-removing.)

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NMS keeps the best box in each overlap cluster — but low τ can wrongly merge two nearby objects.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/object-detection

Start from a pile of overlapping candidate boxes with scores, then step NMS — sort, keep the top box, suppress its high-IoU neighbours — and sweep τ to see over- vs under-suppression.

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Q. Two **distinct** objects of the same class stand very close together. What can NMS with a low τ do wrong?

Summary

Tasks and their losses

Task	Loss
classification	cross-entropy
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Every step adds a term, never removes one: detection is classification **plus** “is anything here?” **plus** “where exactly?”

The detector families

Detector	Key idea	Stages
R-CNN	region proposals → CNN per crop	two (slow)
Fast R-CNN	shared conv + RoI pooling	two
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all produce scored candidate boxes, but the candidate mechanism differs: hand-crafted proposals, learned proposals, or dense cells/anchors. NMS is the common post-processing pattern here.

Final mental model — one idea

Classification answers **what?**

Detection answers **what + where**,
for a **variable set** of objects.

boxes + scores from proposals or dense candidates, cleaned up by NMS

Classification = **what**. Detection = **what + where**, as a set of boxes.

Next (**Lecture 10**): push to **every pixel** — **semantic and instance segmentation**, where the answer is a label map, not a box.